Up, up and away – a hot air story

The story of the invention of the hot air balloon begins in the small town of Alchemy in southern France. Two brothers, Joseph and Jacques Montgolfier, were in the paper manufacturing business. They noticed that whenever they burnt paper, the ashes floated up into air. They discovered that the smoke and heat from the burning paper had a lifting quality.

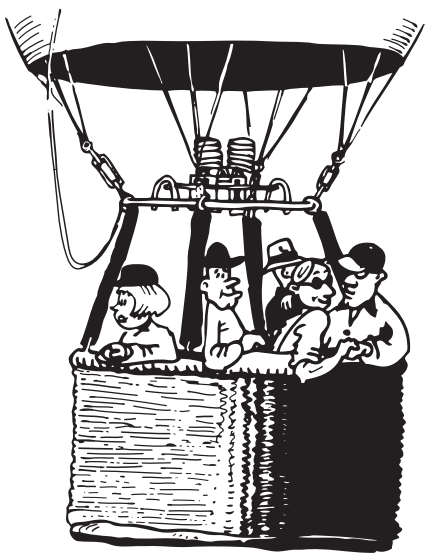
With this in mind, the brothers set out to design a craft that could capture the heat and smoke to allow the craft to lift off the ground. On 19th September 1783, the Montgolfier brothers were successful in creating and launching their invention: a hot air balloon.

Their hot air balloon was made from paper and cloth. They used a combination of chopped wool, straw and dried horse manure for burning in order to inflate the balloon. The straw provided the heat which heated the air inside the balloon. The wool and manure made lots of smoke and kept the flame low so the paper and cloth balloon did not catch fire.

The brothers were a little unsure how the balloon would fly or how safe it was, so they sent sheep, chicken and ducks up in their balloon. Their trials were a success – they managed to keep the animals floating for eight minutes and then land the balloon successfully.

On 21st November 1783, the Montgolfier brothers convinced the King of France to come and see their invention. This time, people would be on board instead of farm animals. Two of their best friends volunteered for the job. The balloon was successfully launched and rose about 150 m above the rooftops of Paris and successfully landed in a vineyard a few kilometres away.

Today, hot air balloons come in all shapes and sizes. The envelope that contains the heated air is made of nylon fabric. The mouth of the balloon which is closer to the flame is made of a fire-resistant material called Nomex. Attached to the envelope is a basket known as the gondola which carries the passengers. Connected above the basket is a burner which injects a flame into the envelope to heat the air. A liquefied gas called propane is stored in a pressure vessel and fuels the burner. At the top of the envelope is a small vent that can be opened briefly while in flight to start a rapid descent. For a slower descent, the air in the balloon is allowed to cool.



So why does the balloon float? Increasing the air temperature inside the envelope makes it lighter than the surrounding air. The balloon floats because of a buoyant force – it is a similar situation to an object floating in water. On a hot day, the balloon must not be loaded as much as it could be on a cool day, because more heat is needed to launch the balloon and too much extra heat could affect the nylon envelope.

Answer the comprehension questions, writing full sentences which contain the question as well as the answer.

On the lines

1 Where did the Montgolfier brothers live?

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2 What kind of business did they run?

_____

_____

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3 What inspired them to make a hot air balloon?

4 What did they send up in their hot air balloons while they were testing them?

Between the lines

5 Use the information in the text to compare and contrast the modern hot air balloons with the Montgolfier brothers' hot air balloon. Summarise your findings in the table below.

Features of the balloon	Montgolfier brothers' hot air balloon	Modern hot air balloons
Shape of the balloon		
Material the envelope is made from		
Fuel used to heat the air		
Other features		

Beyond the lines

6 Why does the air inside the balloon get lighter when it is heated?

7 Explain why the balloon must not be overloaded on a hot day.

8 Modern hot air balloons have a small vent at the top. For a rapid descent, the air vent is opened. Use the particle theory of matter to explain why a rapid descent occurs. Include a labelled diagram in your answer.

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